

EXPERIENCE LAB INTERNSHIP PROGRAMME

Week 6: Project Summary

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HealthConnect Clinic Experience Lab*Data Science Track — Model Improvement, Error Analysis & Validation*

Programme	AnalystLab Africa Experience Lab
Week	Week 6
Project	HealthConnect Clinic Experience Lab
Track	Data Science

1. What Was Planned

The plan was to analyse the weaknesses of the Week 5 baseline model, conduct error analysis, refine features where justified, develop and compare at least one improved model, and complete a genuine cross-track integration with the Data Analytics track.

2. What Was Completed

- Conducted error analysis on the Week 5 baseline, profiling false negatives and false positives separately.
- Refined the `booking_lead_category` feature from three bins to four, using a validated finding from the Data Analytics track.
- Built and evaluated two alternative models (refined Logistic Regression, Random Forest) against the Week 5 baseline.
- Selected a final candidate model based on business-relevant criteria, not just raw accuracy.
- Completed and documented a full cross-track integration with the Data Analytics track.

3. What Was Improved From Week 5

The booking lead time feature was refined from three to four bins using externally validated boundaries. Two alternative models were tested against the baseline, and a documented, criteria-based model selection replaced Week 5's single-model approach.

4. What Was Integrated

A validated no-show rate breakdown by booking lead time, received from the Data Analytics track, was integrated directly into feature engineering. The refined feature's independently recalculated no-show rates matched the shared finding exactly, providing mutual cross-track validation.

5. Track(s) Collaborated With

Data Analytics (primary, formally documented integration). A secondary contact with the Machine Learning Engineering track was also established for future model handoff, to be formalised in Week 7.

6. What Was Exchanged

Received: a validated four-range no-show rate breakdown by booking lead time. Provided: the strongest predictors identified from the Week 5 baseline model (booking lead time, prior no-show history, reminder status), to inform the collaborator's own analytical focus.

7. What Changed as a Result

The feature engineering decision in this notebook was evidence-based rather than self-referential. While the refinement alone did not meaningfully change Logistic Regression's performance, it improved the defensibility of the feature and directly supported the error analysis findings.

8. Key Findings or Development Outcomes

- False negatives cluster among patients with short lead time and low prior no-show rate, patients who look low-risk but are not.
- False positives cluster among patients with long lead time and prior history, showing the model over-relies on these two signals.
- Random Forest offered a small accuracy/precision gain but a slight recall decrease compared to the baseline; given HealthConnect's use case, recall was prioritized, so the refined Logistic Regression was selected as the final candidate.

9. Major Challenges

The clearest challenge was that neither tested improvement (refined feature, alternative model) produced a large performance gain, which required a genuine, undecorated assessment rather than overstating a marginal result.

10. Important Decisions

Chose to prioritize recall over accuracy/precision when selecting the final candidate model, based on the relative business cost of a missed no-show (a wasted slot) versus a false alarm (an unnecessary reminder).

11. Remaining Issues

- No hyperparameter tuning has been performed on Random Forest.
- Performance estimates rely on a single train/test split rather than cross-validation.
- Classification threshold has not yet been adjusted from the default 0.5.

12. Contribution to the Overall HealthConnect Project

This work provides a validated, cross-track-informed no-show prediction model and a clear, evidence-based account of its current limitations, giving the project a realistic foundation for the Machine Learning Engineering track's pipeline integration and the Week 7 testing phase.

13. Proposed Focus for Week 7

Patient-aware cross-validation, Random Forest hyperparameter tuning, feature importance review, classification threshold testing, and formal validation of the candidate model's input/output interface with the ML Engineering track.